VQE

The Variational Quantum Eigensolver

A step-by-step chapter: from eigenvalue problems to a CUDA-Q implementation

What you will be able to do after this chapter

- solve small eigenvalue problems, compute Rayleigh quotients and check answers with trace and determinant;
- set up an electronic-structure problem: Born–Oppenheimer, orbitals, determinants, Hartree–Fock, FCI and correlation;
- use second quantisation and map fermions to qubits with Jordan–Wigner and Bravyi–Kitaev;
- decompose Hamiltonians into Pauli strings, evaluate them on states and plan their measurement and shot budget;
- prove and apply the variational principle, design ansätze (Hartree–Fock, UCC, hardware-efficient) and run the optimisation loop;
- read and modify `VQE.ipynb`, and interpret its results, shot requirements and noise honestly;
- solve every type of question in the accompanying 80-question VQE assessment.

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Companion material in the same repository: `presentations/VQE_Lecture.pdf` (slides), `VQE.ipynb` (CUDA-Q notebook) and `assessments/` (question paper and answer book).

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VQE — The Variational Quantum Eigensolver

How to use this chapter

The chapter is built as a staircase. Sections 1–3 are the *mathematics and chemistry* you need (eigenvalue problems, molecular Hamiltonians, second quantisation). Sections 4–5 are the *quantum-computing toolkit* (fermion-to-qubit mappings, Pauli strings, measurement). Sections 6–9 are *VQE itself* (variational principle, ansätze, optimisation, fully worked problems). Sections 10–12 cover the *implementation* in `VQE.ipynb`, shots, noise and limitations. Section 13 collects problem-solving strategies and maps every assessment question to the section that teaches it; Section 14 has practice problems with answers.

Each section has: the idea in words, the precise statement, a derivation, **worked examples** solved step by step, **common traps** (the wrong answers examiners like to offer) and a short checklist. Statements are marked as *general theory*, *the notebook's implementation*, or *illustrative examples*; "previously recorded" results are the outputs stored in `VQE.ipynb`. Every number was recomputed with NumPy, PySCF/OpenFermion or CUDA-Q (`assessments/vqe_test_source/verify_answers.py`).

Definition: Conventions used everywhere in this chapter

- Pauli words: character k acts on qubit k; basis states $|q_0 q_1 \ldots\rangle$ are written **qubit 0 first**.
- $R_Y(\theta) = e^{-i\theta Y/2}$, $R_Z(\theta) = e^{-i\theta Z/2}$, $S = \mathrm{diag}(1, i)$; CUDA-Q's `exp_pauli(theta, q, word)` $= e^{+i\theta P}$.
- Jordan–Wigner with $|1\rangle$ = occupied: $a_j^\dagger = Z_0 \cdots Z_{j-1} \frac{1}{2}(X_j - iY_j)$, $n_j = \frac{1}{2}(I - Z_j)$.
- Energies of molecules in Hartree (Ha); 1 Ha = 627.51 kcal/mol = 27.211 eV; chemical accuracy 1.6 mHa ≈ 1 kcal/mol.
- The notebook's H_2: 4 qubits, 15 Pauli terms (STO-3G, Jordan–Wigner, spin-orbitals $\sigma_g\alpha, \sigma_g\beta, \sigma_u\alpha, \sigma_u\beta$, bond length 0.740 Å); its default problem is the Ising chain $H = -J\sum Z_i Z_{i+1} - h\sum X_i$ with $N = 15$, $J = 1$, $h = 0.5$.

1 Eigenvalue problems

1.1 Hermitian matrices

Physical energies are eigenvalues of a *Hermitian* operator $H = H^\dagger$. Three facts carry the whole chapter:

Key result: spectral theorem for Hermitian matrices

1. Eigenvalues are real: if $H|v\rangle = \lambda|v\rangle$, $\|v\| = 1$, then $\lambda = \langle v|H|v\rangle = \langle v|H|v\rangle^* = \lambda^*$.
2. Eigenvectors of different eigenvalues are orthogonal: $\mu\langle u|v\rangle = \langle u|H|v\rangle = \lambda\langle u|v\rangle \Rightarrow (\mu - \lambda)\langle u|v\rangle = 0$.
3. There is an orthonormal eigenbasis: $H = \sum_k E_k |E_k\rangle\langle E_k|$, with $E_0 \le E_1 \le \dots$

The smallest eigenvalue E_0 is the **ground-state energy**; its eigenvector is the **ground state**. Real symmetric matrices are Hermitian, so they never have complex eigenvalues (non-symmetric real matrices, such as rotations, can).

1.2 Solving small problems

For a 2×2 matrix $\left(\begin{smallmatrix} a & b \\ b^* & c \end{smallmatrix}\right)$: $\lambda_\pm = \frac{a+c}{2} \pm \sqrt{\left(\frac{a-c}{2}\right)^2 + |b|^2}$. Block-diagonal matrices split into independent smaller problems.

Worked example 1.1: a 2×2 ground state

$H = \left(\begin{smallmatrix} 1 & 2 \\ 2 & -2 \end{smallmatrix}\right)$.

Step 1. $\det(H - \lambda I) = (1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6 = (\lambda+3)(\lambda-2)$: $E_0 = -3$, $E_1 = 2$.

Step 2. $(H + 3I)v = 0$: $4v_1 + 2v_2 = 0$, so $v \propto (1, -2)$; normalised $(1, -2)/\sqrt{5}$.

Step 3. Check: $Hv = (-3, 6)/\sqrt{5} = -3v$. The other eigenvector $(2, 1)/\sqrt{5}$ is orthogonal to it.

Worked example 1.2: a block-diagonal 3×3 matrix

$M = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 5 \end{smallmatrix}\right)$: the block $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has eigenvalues 2 ± 1, the block (5) gives 5. Spectrum $\{1, 3, 5\}$. Checks: $\operatorname{tr} M = 9 = 1 + 3 + 5$ and $\det M = (4 - 1) \cdot 5 = 15 = 1 \cdot 3 \cdot 5$.

1.3 The Rayleigh quotient

Key result: Rayleigh quotient bound

For every non-zero v: $E_0 \le R(v) = \dfrac{v^\dagger H v}{v^\dagger v} \le E_{\max}$, with equality exactly at eigenvectors. If $\|v\| = 1$, $R(v) = \langle v|H|v\rangle$.

Proof. Write $v/\|v\| = \sum_k c_k |E_k\rangle$; then $R(v) = \sum_k |c_k|^2 E_k$, a weighted average of eigenvalues.

Worked example 1.3: Rayleigh quotient of a trial vector

Same H, $v = (1, -1)$: $Hv = (-1, 4)$, $v^\dagger H v = -1 - 4 = -5$, $v^\dagger v = 2$, $R = -2.5$. Weight of the ground vector: $|\langle v, u_0\rangle|^2/\|v\|^2 = (3/\sqrt{5})^2/2 = 0.9$, so $R = 0.9(-3) + 0.1(2) = -2.5$ ✓. The trial energy is *above* E_0.

Common trap

"The ground energy is the smallest diagonal element" — false whenever off-diagonal couplings exist: here $-3 < -2$. Also: $v^\dagger H v$ of an *un-normalised* vector (-5) can lie below E_0; always

divide by $v^\dagger v$.

1.4 Classical cost

n qubits	$D = 2^n$	state (fp64)	practical method
4	16	256 B	dense diagonalisation
15	32 768	524 kB	sparse Lanczos (`eigsh`)
30	1.1×10^9	17 GB	large clusters only
50	1.1×10^{15}	18 PB	impossible

The notebook diagonalises sparse matrices up to `REFERENCE_MAX_QUBITS = 18`; for its Ising chain it uses an exact free-fermion formula that works for any size (Section 10).

Checklist

Real eigenvalues? Orthogonal eigenvectors? Trace and determinant consistent? Rayleigh quotient divided by $v^\dagger v$?

2 Quantum chemistry essentials

2.1 The Schrödinger equation

$i\hbar\,\partial_t |\psi(t)\rangle = H|\psi(t)\rangle$ has the solution $|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$. On an eigenstate this is $e^{-iEt/\hbar}|\psi_E\rangle$: only a *global* phase changes, so all probabilities are constant — a **stationary state**. Energies therefore come from the time-independent equation $H|\psi\rangle = E|\psi\rangle$.

Worked example 2.4: a superposition is not stationary

$|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|E_0\rangle + |E_1\rangle)$ evolves to $\frac{1}{\sqrt{2}}(e^{-iE_0t/\hbar}|E_0\rangle + e^{-iE_1t/\hbar}|E_1\rangle)$. The *relative* phase $e^{-i(E_1-E_0)t/\hbar}$ makes observables that connect the two states oscillate with angular frequency $(E_1 - E_0)/\hbar$; the energy expectation stays $\frac{1}{2}(E_0 + E_1)$.

2.2 The molecular Hamiltonian and Born–Oppenheimer

In atomic units

$$H = -\sum_i \tfrac{1}{2}\nabla_i^2 - \sum_{i,A} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \sum_{A<B} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|}.$$

Nuclei are $\sim 2000\times$ heavier than electrons: fix them (**Born–Oppenheimer**). The last term V_{NN} becomes a number for each geometry; after the qubit mapping it sits in the coefficient of the identity string. Repeating the calculation over geometries gives the potential energy surface (bond lengths, dissociation energies, barriers).

2.3 Units and chemical accuracy

Worked example 2.5: converting an error

An energy error of 0.005 Ha is $0.005 \times 627.51 = 3.14$ kcal/mol — more than three times the chemical-accuracy threshold of 1 kcal/mol = 1/627.51 Ha = 1.594 mHa. The notebook prints "Chemical accuracy (1.6 mHa): REACHED/not reached" for H_2.

2.4 Orbitals, determinants, Hartree–Fock and FCI

Definition: Spin-orbitals and determinants

A basis of N_{orb} spatial orbitals gives $N = 2N_{\text{orb}}$ spin-orbitals (spin α and β). A Slater determinant is fixed by which N_e spin-orbitals are occupied: $\binom{N}{N_e}$ determinants. **Hartree–Fock** (HF) is the best single determinant; **full configuration interaction** (FCI) diagonalises H over all determinants (exact in the basis). $E_{\text{corr}} = E_{\text{FCI}} - E_{\text{HF}} \leq 0$.

Worked example 2.6: H_2 in STO-3G (the notebook's molecule)

Step 1. Orbitals: one $1s$ function per atom $\to$ 2 spatial orbitals $\sigma_g, \sigma_u \to 4$ spin-orbitals; 2 electrons.

Step 2. Determinants: $\binom{4}{2} = 6$; with one α and one β electron: $2 \times 2 = 4$.

Step 3. Energies (from the notebook's Hamiltonian): $E_{\text{HF}} = -1.116759$ Ha, $E_{\text{FCI}} = -1.137284$ Ha, $E_{\text{corr}} = -20.5$ mHa ≈ -12.9 kcal/mol.

Worked example 2.7: LiH in STO-3G (illustration, not in the notebook)

Li: $1s, 2s, 2p_x, 2p_y, 2p_z$; H: $1s \to 6$ spatial orbitals, 12 spin-orbitals $\to$ 12 qubits; 4 electrons. PySCF + OpenFermion at 1.595 Å: 631 Pauli terms, $E_{\text{HF}} = -7.862$ Ha, $E_{\text{FCI}} = -7.882$ Ha. Freezing the Li $1s$ core removes 2 qubits.

Common trap

Restricted HF fails at bond breaking: for H_2 at large R the closed-shell determinant contains 50 % ionic H^+H^- character and lies far too high (at 2.6 Å the gap to FCI is 244 mHa). The nuclear repulsion is *not* the reason — it is identical in every method.

Checklist

Count spatial orbitals, double for spin; include V_{NN} in total energies; correlation energies are negative; convert with 627.51.

3 Second quantisation

Creation and annihilation operators obey the canonical anticommutation relations (CAR)

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \qquad \{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0.$$

Key result: consequences of the CAR

$(a_p^\dagger)^2 = 0$ (Pauli exclusion, from $\{a_p^\dagger, a_p^\dagger\} = 0$); $n_p = a_p^\dagger a_p$ satisfies $n_p^2 = a_p^\dagger(1 - a_p^\dagger a_p)a_p = n_p$, so its eigenvalues are 0, 1; $a_p |\text{vac}\rangle = 0$ (the zero vector).

The electronic Hamiltonian becomes

$$H = \sum_{pq} h_{pq} a_p^\dagger a_q + \tfrac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s + E_{\text{const}},$$

with one-electron integrals h_{pq} (kinetic + nuclear attraction) and electron-repulsion integrals h_{pqrs} computed classically (PySCF). It has $O(N^4)$ terms (symmetries reduce the count by about 8, not the power) — polynomial, although the state space is 2^N-dimensional.

Common trap

Distinguish the zero *operator* $(a_p^\dagger)^2$ from the zero *vector* $a_p |\text{vac}\rangle$; and $O(N^4)$ terms from the 2^N-dimensional state space.

4 From fermions to qubits

4.1 Jordan–Wigner

One qubit per spin-orbital, $|1\rangle$ = occupied. Then $\frac{1}{2}(X - iY) = |1\rangle\langle 0|$ fills an orbital, and the string of Z's on all earlier qubits produces the sign $(-1)^{n_0 + \cdots + n_{j-1}}$ that makes operators on different orbitals anticommute:

$$a_j^\dagger = Z_0 \cdots Z_{j-1} \tfrac{1}{2}(X_j - iY_j), \qquad n_j = \tfrac{1}{2}(I - Z_j).$$

Worked example 4.8: $a_2^\dagger$ on 4 qubits

$a_2^\dagger = \frac{1}{2} Z_0 Z_1 (X_2 - iY_2) = \frac{1}{2}$ `ZZXI` $- \frac{i}{2}$ `ZZYI` (OpenFermion gives exactly these coefficients). Its Pauli weight is 3: the weight of $a_j^\dagger$ is $j + 1$, i.e. $O(N)$.

Worked example 4.9: the hopping term, derived

Step 1. $a_0^\dagger = \frac{1}{2}(X_0 - iY_0)$, $a_1 = Z_0 \frac{1}{2}(X_1 + iY_1)$.

Step 2. $(X - iY)Z = XZ - iYZ = -iY + X$, so $(X_0 - iY_0)Z_0 = X_0 - iY_0$.

Step 3. $a_0^\dagger a_1 = \frac{1}{4}(X_0 - iY_0)(X_1 + iY_1) = \frac{1}{4}(X_0X_1 + Y_0Y_1 + iX_0Y_1 - iY_0X_1)$.

Step 4. Adding the conjugate $a_1^\dagger a_0$ cancels the imaginary parts: $a_0^\dagger a_1 + a_1^\dagger a_0 = \frac{1}{2}(X_0X_1 + Y_0Y_1)$.

It swaps $|01\rangle \leftrightarrow |10\rangle$ and annihilates $|00\rangle$, $|11\rangle$. ($\frac{1}{2}(X_0X_1 - Y_0Y_1)$ would be the pairing term $a^\dagger a^\dagger + aa$.)

Worked example 4.10: a density–density term

$n_0 n_1 = \frac{1}{4}(I - Z_0)(I - Z_1) = \frac{1}{4}(I - Z_0 - Z_1 + Z_0Z_1)$ — this is how the Z-type terms of the H_2 Hamiltonian arise.

Worked example 4.11: a double excitation

$a_3^\dagger a_2^\dagger a_1 a_0 - \text{h.c.}$ becomes 8 commuting Pauli strings with coefficients $\pm i/8$ (`XXXY`, `XXYX`, `XYXX`, `YXXX`, `XYYY`, ...; each with an odd number of Y's). The notebook's UCC kernel keeps only `XXXY` (Section 7).

4.2 Parity, Bravyi–Kitaev and tapering

- **Parity mapping**: qubit j stores $n_0 \oplus \cdots \oplus n_j$; parities become local, occupations non-local.
- **Bravyi–Kitaev**: qubits store partial sums on a binary tree, $b = Bn \bmod 2$; Pauli weight $O(\log N)$ (for $a_7^\dagger$ on 8 modes: 4 instead of 8).
- **Tapering**: $\mathbb{Z}_2$ symmetries (electron-number and spin parity) remove qubits exactly; H_2 goes from 4 to 2 qubits.

Worked example 4.12: Bravyi–Kitaev encoding of Hartree–Fock

$B = \left(\begin{smallmatrix} 1&0&0&0 \\ 1&1&0&0 \\ 0&0&1&0 \\ 1&1&1&1 \end{smallmatrix}\right)$, $n = (1,1,0,0)$: $b_0 = 1$, $b_1 = 1+1 \equiv 0$, $b_2 = 0$, $b_3 = 2 \equiv 0 \Rightarrow |1000\rangle$. Check: the Bravyi–Kitaev H_2 Hamiltonian has $\langle 1000|H_{\text{BK}}|1000\rangle = E_{\text{HF}}$.

Common trap

The same physical state has different bitstrings in different encodings ($|1100\rangle$ in Jordan–Wigner, $|1000\rangle$ in Bravyi–Kitaev). Preparing the Jordan–Wigner reference with a Bravyi–Kitaev Hamiltonian gives a wrong energy.

4.3 The notebook's H_2 qubit Hamiltonian

15 terms: the identity (-0.097066, including V_{NN}), 10 Z-type terms and 4 exchange terms ± 0.045303 (`XXYY`, `YYXX` negative; `XYYX`, `YXXY` positive). PySCF + OpenFermion reproduce every coefficient to 10^{-15} at $R = 0.740$ Å (the notebook's text originally said 0.7414 Å, where the coefficients would differ by up to 1.8 mHa; it has been corrected).

Checklist

$|1\rangle$ = occupied; Z-string on *earlier* qubits; $\frac{1}{2}(X - iY)$ creates; check anticommutation on a small example; re-encode references.

5 Pauli strings and measurement

5.1 Pauli decomposition

The 4^n Pauli strings are an orthogonal basis of $2^n \times 2^n$ matrices, $\text{tr}(PQ) = 2^n \delta_{PQ}$. Hence

$$H = \sum_P c_P P, \qquad c_P = 2^{-n} \,\text{tr}(PH) \in \mathbb{R} \text{ for Hermitian } H.$$

Worked example 5.13: a 2×2 Hermitian matrix

$M = \left(\begin{smallmatrix} 3 & 1-2i \\ 1+2i & -1 \end{smallmatrix}\right)$: $c_I = \frac{3-1}{2} = 1$, $c_Z = \frac{3+1}{2} = 2$, $c_X = \operatorname{Re} M_{01} = 1$, and $M_{01} = c_X - ic_Y$ gives $c_Y = 2$. So $M = I + X + 2Y + 2Z$.

Worked example 5.14: a diagonal two-qubit matrix

$\operatorname{diag}(1, 2, 3, 4)$: $c_{II} = 2.5$, $c_{ZI} = \frac{1+2-3-4}{4} = -1$, $c_{IZ} = \frac{1-2+3-4}{4} = -0.5$, $c_{ZZ} = 0$. Check $|11\rangle$: $2.5 + 1 + 0.5 = 4$.

5.2 Pauli algebra and exponentials

$XY = iZ$, $YZ = iX$, $ZX = iY$ (reverse order: minus sign). Strings multiply position by position; two strings commute iff they anticommute at an *even* number of positions. For $P^2 = I$: $e^{i\theta P} = \cos\theta\, I + i \sin\theta\, P$.

Worked example 5.15: products and actions

$(X \otimes Y)(Y \otimes X) = (iZ) \otimes (-iZ) = Z \otimes Z$. $\quad X_0X_1X_2Y_3\,|1100\rangle = |0\rangle\,|0\rangle\,|1\rangle\,(i\,|1\rangle) = i\,|0011\rangle$. $e^{i\theta Z\otimes Z}\,|01\rangle = e^{-i\theta}\,|01\rangle$ (eigenvalue -1).

5.3 Measuring Pauli strings

Hardware measures Z. Rotate each factor first: X: H (since $HXH = Z$); Y: $S^\dagger$ then H (since $(HS^\dagger)Y(HS^\dagger)^\dagger = HXH = Z$); Z: nothing. A shot with bits b gives the outcome $(-1)^{\sum_{k\in \operatorname{supp} P} b_k}$ of P.

Key result: shot statistics

A ± 1 observable has variance $1 - \langle P\rangle^2 \le 1$: standard error $\le 1/\sqrt{S}$. An energy estimator with per-shot spread σ needs $S = (\sigma/\varepsilon)^2$ shots for precision ε. $\operatorname{Var}(H) = \langle H^2\rangle - \langle H\rangle^2 = 0$ exactly for eigenstates (zero-variance principle).

Worked example 5.16: measurement settings for H_2

All 10 Z-type terms are qubit-wise commuting: one setting. No two of the four exchange strings are qubit-wise commuting (X meets Y somewhere), so they need 4 settings: **5** in total with qubit-wise grouping. The four exchange strings commute pairwise, so with general commuting groups (one entangling basis change) they share one setting: **2** in total.

Worked example 5.17: a shot budget

Per-shot spread $\sigma = 0.1$ Ha, target 1.6 mHa: $S = (0.1/0.0016)^2 = 3906$ shots per energy evaluation. CUDA-Q simulation of the optimal H_2 state gives a spread of the 15-term estimate of a few mHa at 1000 shots and about one mHa at 10 000 shots (lecture figure).

Common trap

Qubit order: the notebook writes qubit 0 first, `cudaq.get_state` lists qubit 0 as the *least* significant bit — $|1100\rangle$ is index 3, not 12. And: $\langle\psi|H|\psi\rangle$ is real for every state because $H = H^\dagger$; normalisation has nothing to do with it.

Checklist

$c_P = 2^{-n}\,\mathrm{tr}(PH)$; count anticommuting positions; basis change before measuring X/Y; shots scale as $1/\varepsilon^2$.

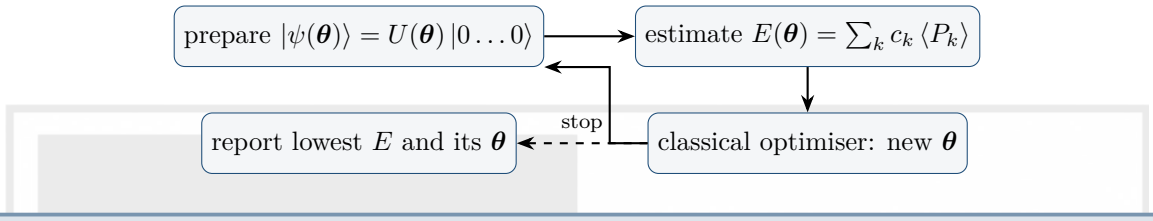
6 The variational principle and the VQE loop

Key result: variational principle

For every normalised $|\psi\rangle$: $\langle\psi|H|\psi\rangle = \sum_k |c_k|^2 E_k \geq E_0$, with equality iff $|\psi\rangle$ is a ground state. For every density matrix: $\mathrm{tr}(\rho H) = \sum_j p_j \langle\psi_j|H|\psi_j\rangle \geq E_0$ — the bound survives noise.

Key result: Rayleigh–Ritz (Hylleraas–Undheim–MacDonald)

Diagonalising H inside an m-dimensional subspace gives Ritz values $\varepsilon_k \geq E_{k-1}$, $k = 1, \ldots, m$; enlarging the subspace can only lower them.



Worked example 6.18: one qubit, solved completely

$H = -Z + \frac{1}{2}X$, $|\psi(\theta)\rangle = R_Y(\theta)\,|0\rangle$.

Step 1. $\langle Z\rangle = \cos\theta$, $\langle X\rangle = \sin\theta$, so $E(\theta) = -\cos\theta + \frac{1}{2}\sin\theta$.

Step 2. $E'(\theta) = \sin\theta + \frac{1}{2}\cos\theta = 0$: $\tan\theta = -\frac{1}{2}$; the minimum is $\theta^* = -\arctan\frac{1}{2} = -0.4636$ ($E'' = \cos\theta^* - \frac{1}{2}\sin\theta^* > 0$).

Step 3. $E(\theta^*) = -0.894 - 0.224 = -1.118 = -\sqrt{5}/2 = E_0$: the one-parameter ansatz is exact (a real ground state of a real 2×2 matrix).

Step 4. Start $\theta = 0$: $E = -1$, error 0.118; variance $\langle H^2\rangle - \langle H\rangle^2 = 1.25 - 1 = 0.25$ ($H^2 = \frac{5}{4}I$), zero at θ^*.

Worked example 6.19: contamination costs energy linearly

90 % ground state (-1.13728 Ha) and 10 % of an eigenstate at -0.53821 Ha: $E = -1.0774$ Ha, error $0.1 \times 0.59907 = 59.9$ mHa.

Common trap

An energy below E_0 computed with exact expectation values is a bug (Hamiltonian, sign, reference or normalisation), not a discovery. With finite shots single estimates *can* dip below E_0, and error-mitigated (extrapolated) energies carry no variational guarantee.

Checklist

Normalised state? Exact or sampled energy? Report the best evaluated point and compare with a reference.

7 Ansatz design

7.1 Hartree–Fock reference

Under Jordan–Wigner the HF determinant is a basis state, prepared with X gates (H_2: `x(qubits[0]); x(qubits[1])` $\to |1100\rangle$).

Worked example 7.20: the HF energy term by term

On $|1100\rangle$: $Z_0 = Z_1 = -1$, $Z_2 = Z_3 = +1$; strings with X/Y contribute 0.

term	c_k	value on $\|1100\rangle$	contribution
`IIII`	−0.097066	+1	−0.097066
`ZIII`, `IZII`	+0.171413	−1	−0.342826
`IIZI`, `IIIZ`	−0.223432	+1	−0.446863
`ZZII`	+0.168689	+1	+0.168689
`ZIZI`, `IZIZ`	+0.120625	−1	−0.241250
`ZIIZ`, `IZZI`	+0.165928	−1	−0.331856
`IIZZ`	+0.174413	+1	+0.174413
total			−1.116759

7.2 Unitary coupled cluster

$|\psi\rangle = e^{T-T^\dagger}|\mathrm{HF}\rangle$, $T = \sum_{ia} t_i^a a_a^\dagger a_i + \sum_{ijab} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i$ (i,j occupied, a,b virtual). $T - T^\dagger$ is anti-Hermitian, so the operator is unitary. Counts (spin-orbitals, no symmetry): singles $n_o n_v$, doubles $\binom{n_o}{2}\binom{n_v}{2}$.

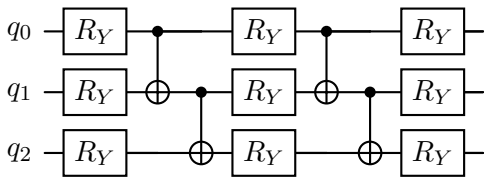
Worked example 7.21: counting excitations

H_2: $n_o = n_v = 2$: 4 singles, 1 double. LiH (STO-3G): $n_o = 4$, $n_v = 8$: $4 \cdot 8 = 32$ singles, $\binom{4}{2}\binom{8}{2} = 6 \cdot 28 = 168$ doubles.

Key result: the notebook's UCC ansatz for H_2

$|\psi(\theta)\rangle = e^{i\theta X_0 X_1 X_2 Y_3}|1100\rangle = \cos\theta\,|1100\rangle + i\sin\theta\,(i\,|0011\rangle) = \cos\theta\,|1100\rangle - \sin\theta\,|0011\rangle$. The exact ground state lies in this plane ($0.9937\,|1100\rangle - 0.1125\,|0011\rangle$; single excitations vanish by symmetry), so one parameter reaches FCI.

7.3 Hardware-efficient ansatz



L layers of [R_Y on all qubits, CNOT ladder $q \to q+1$] plus a final R_Y layer: $(L+1)N$ parameters, $L(N-1)$ CNOTs.

Worked example 7.22: counting the hardware-efficient circuit

Default $N = 15$, $L = 2$: 45 parameters, 28 CNOTs, 73 gates (recorded statistics `ry 45, cx 28`, depth 20). $N = 10$, $L = 3$: 40, 27, 67.

Only real gates are used, so all amplitudes are real — adequate because the Ising and H_2 Hamiltonians are real symmetric (a real ground state exists). Initial angles $\sim \mathcal{N}(0, 0.1^2)$ start near $|0 \dots 0\rangle$, whose Ising energy is $-J(N-1) = -14$.

Common trap

Deep, random hardware-efficient circuits suffer from *barren plateaus*: gradient variances shrink exponentially with n. Structured (chemistry-inspired) ansätze, Hartree–Fock starts and local cost functions help.

Checklist

Reference state in the right encoding? Ansatz contains the ground state (or how far from it)? Parameter and CNOT counts?

8 Classical optimisation

optimiser	principle	cost per step
COBYLA (default)	linear models in a shrinking trust region	1 evaluation
Nelder–Mead	simplex of $n_p + 1$ points	1–2 evaluations
Powell	line searches along conjugate directions	several evaluations
L-BFGS-B	quasi-Newton, finite-difference gradients	$n_p + 1$ evaluations

Key result: parameter-shift rule

For a gate $e^{-i\theta P/2}$, $E(\theta) = a + b\cos\theta + c\sin\theta$, hence $\partial_\theta E = \frac{1}{2}[E(\theta + \frac{\pi}{2}) - E(\theta - \frac{\pi}{2})]$ exactly: 2 runs per parameter (90 for the default 45-parameter ansatz; finite differences: 46, approximate).

Worked example 8.23: gradient descent by hand

$E(\theta) = -\cos\theta + \frac{1}{2}\sin\theta$, $\theta_0 = 0$, step $\theta_{k+1} = \theta_k - 0.5\,\partial_\theta E$:

k	θ_k	$\partial_\theta E$	$E(\theta_k)$
0	0	0.500	−1.0000
1	−0.250	0.237	−1.0926
2	−0.369	0.106	−1.1130
3	−0.422	0.047	−1.1170

At $k = 0$: $E(\frac{\pi}{2}) = 0.5$, $E(-\frac{\pi}{2}) = -0.5$, so $\partial_\theta E = 0.5$. The energies approach $E_0 = -1.1180$ from above.

In the notebook: `minimize(energy, initial_theta, method="COBYLA", options={"maxiter": 300, "rhobeg": 0.3})`; every evaluation is stored and the *best* one (`argmin(energy_history)`) is reported, not necessarily `opt.x`. A `MAXFUN` message means the budget, not convergence, stopped the run.

9 Fully worked problems

Worked example 9.24: H_2 as a 2×2 problem

Step 1. Matrix elements from the 15 terms: $H_{11} = \langle 1100|H|1100\rangle = -1.116759$ (Section 7), $H_{22} = \langle 0011|H|0011\rangle = 0.462618$, $H_{12} = \langle 1100|H|0011\rangle = 0.181210$ Ha.

Step 2. H_{12} comes only from the exchange strings: e.g. $\texttt{XXYY}\,|0011\rangle = (-i)^2\,|1100\rangle = -\,|1100\rangle$, times $c = -0.045303$ gives $+0.045303$; all four contribute equally: $4 \times 0.045303 = 0.181210$.

Step 3. $E(\theta) = H_{11}\cos^2\theta + H_{22}\sin^2\theta - H_{12}\sin 2\theta$; $E' = 0$: $\tan 2\theta^* = \frac{2H_{12}}{H_{22}-H_{11}} = \frac{0.362420}{1.579377} = 0.2295$, $\theta^* = 0.1128$.

Step 4. $E_{\min} = \frac{H_{11}+H_{22}}{2} - \sqrt{\left(\frac{H_{22}-H_{11}}{2}\right)^2 + H_{12}^2} = -0.327071 - 0.810213 = -1.137284$ Ha $= E_{\text{FCI}}$.

Step 5. Weights: $\cos^2\theta^* = 0.9873$, $\sin^2\theta^* = 0.0127$. The 1.3 % admixture of $|0011\rangle$ is worth 20.5 mHa.

Worked example 9.25: two-qubit Ising chain

$H = -Z_0Z_1 - \frac{1}{2}(X_0 + X_1)$; circuit $R_Y(\theta_0) \otimes R_Y(\theta_1)$ then $\text{CNOT}_{0\to1}$.

Step 1. $\theta_0 = \frac{\pi}{2}$, $\theta_1 = 0$: $R_Y(\frac{\pi}{2})\,|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, CNOT $\to |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

Step 2. $\langle Z_0Z_1\rangle = 1$, $\langle X_0\rangle = \langle X_1\rangle = 0$: $E = -1$.

Step 3. H commutes with X_0X_1; in the block $\{|\Phi^+\rangle, |\Psi^+\rangle\}$: $\left(\begin{smallmatrix}-1 & -1\\ -1 & 1\end{smallmatrix}\right)$, eigenvalues $\pm\sqrt{2}$: $E_0 = -\sqrt{2}$. Relative error 29.3 %.

Step 4. In general, after the CNOT $\langle Z_0Z_1\rangle = \cos\theta_1$ (CNOT maps $Z_0Z_1 \to Z_1$ in the Heisenberg picture).

Worked example 9.26: energy from measured Pauli averages

$H = 0.5\,Z_0 - 0.3\,Z_1 + 0.2\,X_0X_1$ with $\langle Z_0\rangle = 0.6$, $\langle Z_1\rangle = -0.2$, $\langle X_0X_1\rangle = 0.5$: $E = 0.30 + 0.06 + 0.10 = 0.46$. Reminder: $\langle Z\rangle = P(0) - P(1)$, so for example a qubit with $P(1) = 0.2$ has $\langle Z\rangle = 0.6$.

10 The implementation in `VQE.ipynb`

10.1 Structure and code

1. Section 3 (inputs): `HAMILTONIAN = "tfim" | "H2" | "custom"`, `N_QUBITS = 15`, `TFIM_J = 1.0`, `TFIM_H = 0.5`, `ANSATZ = "auto"`, `ANSATZ_LAYERS = 2`, `OPTIMIZER = "COBYLA"`, `MAX_EVALUATIONS = 300`, `OBJECTIVE_SHOTS = None`, `SHOTS = 4000`, `SEED = 42`.
2. Section 4: Pauli terms (Ising: 14 ZZ + 15 X = 29), `build_spin_operator` with `cudaq.SpinOperator.from_word`, exact reference.
3. Section 7: kernels `hea_ansatz`, `ucc_ansatz` (+ sampled versions with `mz`).
4. Section 8: `energy(theta)` = `cudaq.observe(kernel, hamiltonian, *args).expectation()`, COBYLA loop, best-of-history, final `cudaq.sample`.
5. Sections 9–10: convergence/error/bitstring plots, benchmark report in `exports/vqe/`.

Key result: exact Ising reference by free fermions

The open chain maps to free fermions; with a $2N \times 2N$ real antisymmetric matrix A, $E_0 = -\frac{1}{2}\sum|\lambda(iA)|$ for any N ($N = 100$: a 200×200 matrix). Cross-check by sparse diagonalisation for $N \leq 16$; for H_2 and custom Hamiltonians, sparse diagonalisation up to 18 qubits.

Backends: `qpp-cpu`, `density-matrix-cpu`, `nvidia` (fp32/fp64), `tensornet(-mps)`, multi-GPU via MPI, hardware providers; `BACKEND = "gpu"` without a GPU raises a `RuntimeError`. Only the CPU state-vector path has been executed.

10.2 Results

	previously recorded	re-executed notebook	lecture script
E_{VQE}	-15.0533898284	-15.0533898284	-15.0533898284
error vs. -15.0825179883	0.0291 (0.193 %)	same	same
evaluations	300 (budget)	300 (budget)	300
$P(\lvert 0\ldots0\rangle)$	0.7005	0.7005	0.7005

Worked example 10.27: reading the Ising result honestly

Step 1. Relative error $0.0291/15.0825 = 0.193\,\%$ — looks tiny.

Step 2. But $|0\ldots0\rangle$ alone gives -14. Measured from there, VQE recovered $\frac{-14+15.0534}{-14+15.0825} = \frac{1.0534}{1.0825} = 97.3\,\%$ of the remaining energy.

Step 3. The budget ended the run (`MAXFUN`); more evaluations, then more layers, should lower the error.

Step 4. The sampled state is $70\,\%$ $|0\ldots0\rangle$ although H is flip-symmetric: symmetry breaking, at an energy cost exponentially small in N.

$\mathbf{H_2}$ with the UCC ansatz (lecture script, notebook settings): chemical accuracy after 8 evaluations, error 1.3×10^{-10} Ha after 22.

11 Shots, noise and mitigation

- **Complete depolarisation**: $\rho = I/2^n$, $E = \mathrm{tr}(H)/2^n = c_{I\ldots I}$ (-0.0971 Ha for H_2). Noise pulls energies up but keeps $\mathrm{tr}\,\rho H \geq E_0$.
- **Global-depolarising model** (illustration): $E(p) = (1-p)E_{\mathrm{FCI}} + p\,c_{IIII}$; already $p = 0.02$ costs the entire correlation energy of H_2.
- **Zero-noise extrapolation**: $E(\lambda{=}1) = -1.10$, $E(3) = -1.00$, linear $\Rightarrow E(0) = -1.10-0.05 = -1.15$ Ha; can land below E_0.
- **Memory**: state vector $16\cdot2^n$ B (12 qubits: 65.5 kB; 15 qubits: 524 kB), density matrix $16\cdot4^n$ B (12 qubits: 268 MB).

Common trap

"With finite shots VQE energies can never fall below E_0" is false: the *expectation* obeys the bound, individual estimates do not.

12 Limitations and advanced methods

Measurement cost ($O(N^4)$ terms $\times$ $(\sigma/\varepsilon)^2$ shots $\times$ hundreds of evaluations), barren plateaus, local minima, limited expressiveness and noise limit VQE; no chemical quantum advantage has been demonstrated (CCSD(T) and DMRG handle far larger molecules). Compared with quantum phase estimation, VQE needs shallow circuits but $O(1/\varepsilon^2)$ shots instead of $O(1/\varepsilon)$ coherent evolution.

Advanced directions: ADAPT-VQE (operators chosen by gradient), k-UpCCGSD and qubit-coupled cluster, active spaces and tapering, excited states by *variational quantum deflation* ($F = \langle H \rangle + \beta |\langle \psi_0 | \psi \rangle|^2$ with $\beta > E_1 - E_0$), subspace-search VQE and quantum subspace expansion; error mitigation. Applications: reaction energies and barriers, potential energy surfaces, materials and magnetism.

13 Problem-solving strategies and question map

13.1 Strategy by question format

- **Single-correct numerical:** compute fully, then sanity-check: energies above E_0; probabilities in $[0, 1]$; signs of Z on occupied qubits; units (Ha, mHa, kcal/mol). Typical distractors: "amplitude instead of probability", "forgot the constant term", "bit order reversed", "half angle forgotten", "variance instead of standard deviation".
- **Matching lists:** solve the easiest row first (often a pure definition), then eliminate options.
- **Assertion–Reason:** judge A and R *independently*; only if both are true ask whether R *explains* A. True-but-irrelevant reasons (nuclear repulsion, real initial states) are common.
- **Statement I/II:** one counterexample kills a "for every" statement; read code-related statements against the actual code.
- **Paragraph:** the first question sets up a quantity (a matrix, an energy) reused in the second.

13.2 Fast checks that catch most mistakes

1. Is the energy $\geq E_0$? If not, look for the bug.
2. On a basis state, only I/Z strings contribute; occupied qubits give $Z = -1$.
3. Pauli products: track $\pm i$ factor by factor.
4. Shrink matrices with symmetry (2×2 blocks).
5. Recorded vs. re-executed vs. illustrative numbers — which one does the question use?

13.3 Map: assessment questions → sections of this chapter

Questions	Where to learn it
Q.1–Q.3, Q.8, Q.13–Q.15	Section 1 (eigenproblems, Rayleigh quotient, Hermitian spectra) and Section 6
Q.4–Q.6, Q.9, Q.10, Q.18–Q.20	Section 2 (Schrödinger equation, Born–Oppenheimer, determinants, correlation)
Q.11, Q.12	Section 3 (second quantisation)
Q.16, Q.17, Q.21–Q.36, Q.38–Q.40	Section 5 (Pauli strings, measurement, ordering, variance)
Q.7, Q.37, Q.41, Q.42, Q.53, Q.71	Section 6 (variational principle, Rayleigh–Ritz)
Q.43, Q.44, Q.48, Q.49, Q.56, Q.66, Q.67	Section 7 (Hartree–Fock, UCC, hardware-efficient ansatz)
Q.45, Q.47, Q.54, Q.59, Q.60	Sections 5 and 9 (grouping, worked problems)
Q.51, Q.52, Q.58, Q.68	Section 8 (optimisers, gradients, convergence)
Q.61–Q.65	Section 4 (Jordan–Wigner, Bravyi–Kitaev, encodings)
Q.46, Q.50, Q.55, Q.57, Q.73–Q.76, Q.79, Q.80	Section 10 (the notebook, references, backends, results)
Q.69, Q.70, Q.72	Section 11 (shots, noise, mitigation)
Q.77, Q.78	Sections 2 and 4 (LiH: frozen core, Pauli-term count)

(Section numbers: 1 eigenproblems, 2 chemistry, 3 second quantisation, 4 mappings, 5 Pauli strings and measurement, 6 variational principle, 7 ansätze, 8 optimisation, 9 worked problems, 10 notebook, 11 shots and noise, 12 limitations.)

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14 Practice problems

Try each before looking at the answers on the next page. They use the same ideas as the assessment but different numbers.

P1. Eigenvalues of $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and the Rayleigh quotient of $(1, 0)$.

P2. Pauli decomposition of $\left(\begin{smallmatrix} 0 & -i \\ i & 2 \end{smallmatrix}\right)$.

P3. Number of 2-electron determinants in 6 spin-orbitals, and how many have $S_z = 0$.

P4. Jordan–Wigner form of $n_0 n_1$ and of $a_1^\dagger$ on 3 qubits.

P5. $\langle \Phi^+ | X_0 X_1 | \Phi^+ \rangle$ and $\langle \Phi^+ | Z_0 | \Phi^+ \rangle$ for $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

P6. For $H = 2Z + X$ and the trial state $R_Y(\theta)\,|0\rangle$: $E(\theta)$, E_0 and θ^*.

P7. Hardware-efficient counts for $N = 8$, $L = 4$.

P8. $E(\theta)$ for $\cos\theta\,|u\rangle - \sin\theta\,|v\rangle$ with $H_{uu} = 0$, $H_{vv} = 1$, $H_{uv} = -0.5$; its minimum.

P9. Shots for 0.5 mHa precision with a per-shot spread of 0.05 Ha.

P10. Energy of $|0 \ldots 0\rangle$ for the Ising chain with $N = 10$, $J = 2$, $h = 0.3$.

P11. $X_0 Y_1 Z_2\,|011\rangle$.

P12. Bravyi–Kitaev bits for the occupation $(1, 0, 1, 0)$ with the 4×4 matrix B of Section 4.

P13. Singles and doubles for 10 spin-orbitals and 4 electrons.

P14. One parameter-shift gradient step from $\theta = 0$ for $E(\theta) = \cos\theta - \sin\theta$ with step size 0.5: new θ and E.

P15. The fully depolarised energy of $H = -0.5\,II + 0.3\,ZZ - 0.2\,XI$.

P16. A VQE with exact expectation values reports -1.140 Ha for the notebook's H_2. What do you conclude?

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Answers to the practice problems

P1. 1 and 3; $R((1,0)) = 2$ (between them).

P2. $c_I = 1$, $c_Z = -1$, $c_X = 0$, $c_Y = 1$: $I + Y - Z$ (check $M_{01} = c_X - ic_Y = -i$).

P3. $\binom{6}{2} = 15$; $S_z = 0$: $3 \times 3 = 9$.

P4. $n_0 n_1 = \frac{1}{4}(I - Z_0 - Z_1 + Z_0 Z_1)$; $a_1^\dagger = Z_0 \otimes \frac{1}{2}(X_1 - iY_1) \otimes I_2$.

P5. 1 and 0.

P6. $E = 2\cos\theta + \sin\theta = \sqrt{5}\cos(\theta - \varphi)$, $E_0 = -\sqrt{5}$, $\theta^* = \pi + \arctan\frac{1}{2}$ (equivalently $-\pi + 0.4636$).

P7. 40 parameters, 28 CNOTs, 68 gates.

P8. $E = \sin^2\theta + 0.5\sin 2\theta$; eigenvalues of $\left(\begin{smallmatrix} 0 & -0.5 \\ -0.5 & 1 \end{smallmatrix}\right)$ are $\frac{1}{2} \pm \frac{\sqrt{2}}{2}$; minimum -0.2071.

P9. $(0.05/0.0005)^2 = 10\,000$.

P10. $-2 \times 9 = -18$ (the field term has zero expectation).

P11. $X|0\rangle = |1\rangle$, $Y|1\rangle = -i|0\rangle$, $Z|1\rangle = -|1\rangle$: $(-i)(-1)|101\rangle = i|101\rangle$.

P12. $b_0 = 1$, $b_1 = 1 + 0 = 1$, $b_2 = 1$, $b_3 = 1 + 0 + 1 + 0 = 0$: $|1110\rangle$.

P13. $4 \times 6 = 24$ singles; $\binom{4}{2}\binom{6}{2} = 6 \times 15 = 90$ doubles.

P14. $E(\frac{\pi}{2}) = -1$, $E(-\frac{\pi}{2}) = 1$, gradient $= \frac{1}{2}(-1-1) = -1$; $\theta_1 = 0 + 0.5 = 0.5$, $E = \cos 0.5 - \sin 0.5 = 0.398$ (down from 1).

P15. Only the identity survives: -0.5.

P16. It is 2.7 mHa *below* FCI (-1.137284): impossible for a normalised state with exact expectation values — there is a bug.

Summary of the chapter in ten lines

1. Ground energy = smallest eigenvalue of a Hermitian H; Rayleigh quotients bound it from above.
2. Born–Oppenheimer + basis → second-quantised H with $O(N^4)$ terms; constants (incl. V_{NN}) sit in $c_{I\ldots I}$.
3. Jordan–Wigner: occupation per qubit + Z-strings; Bravyi–Kitaev: $O(\log N)$ weight; re-encode reference states.
4. $c_P = 2^{-n}\,\mathrm{tr}(PH)$; measure P after H / $S^\dagger H$ rotations; group commuting terms; shots $\propto 1/\varepsilon^2$.
5. Variational principle: $\langle\psi|H|\psi\rangle \geq E_0$ for every normalised (or mixed) state; Rayleigh–Ritz for subspaces.
6. VQE = circuit state + measured energy + classical optimiser; report the best evaluated point.
7. H_2: HF $|1100\rangle + e^{i\theta XXXY}$ gives FCI exactly ($\theta^* = 0.1128$, 1.3 % of $|0011\rangle$, 20.5 mHa).
8. Hardware-efficient: $(L+1)N$ angles, $L(N-1)$ CNOTs; trainability issues when deep.
9. Noise pulls energies towards $c_{I\ldots I}$; mitigated values lose the variational guarantee.
10. Read results honestly: exact vs. shot energies, budget-limited runs, recorded vs. re-executed vs. illustrative.

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